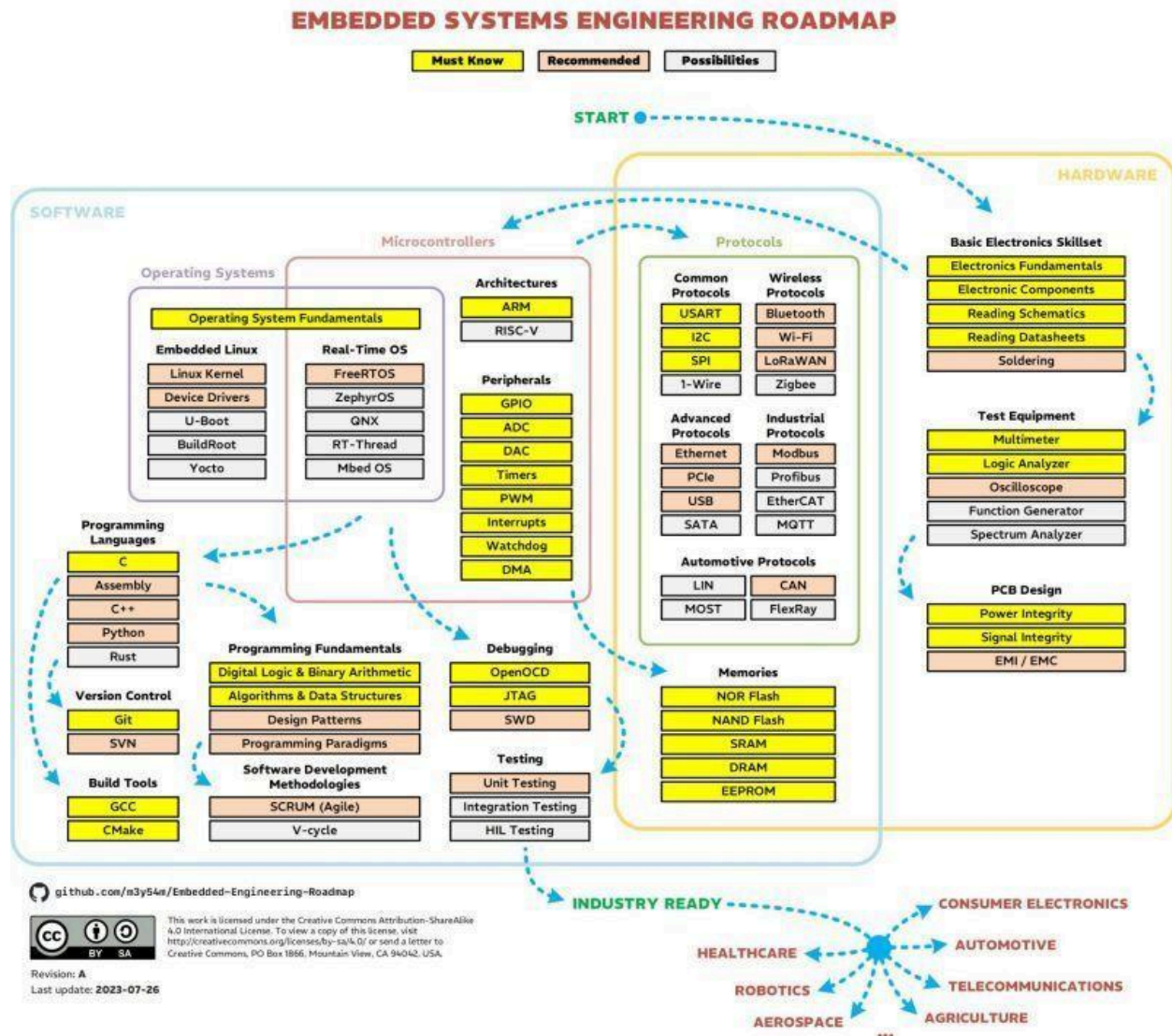
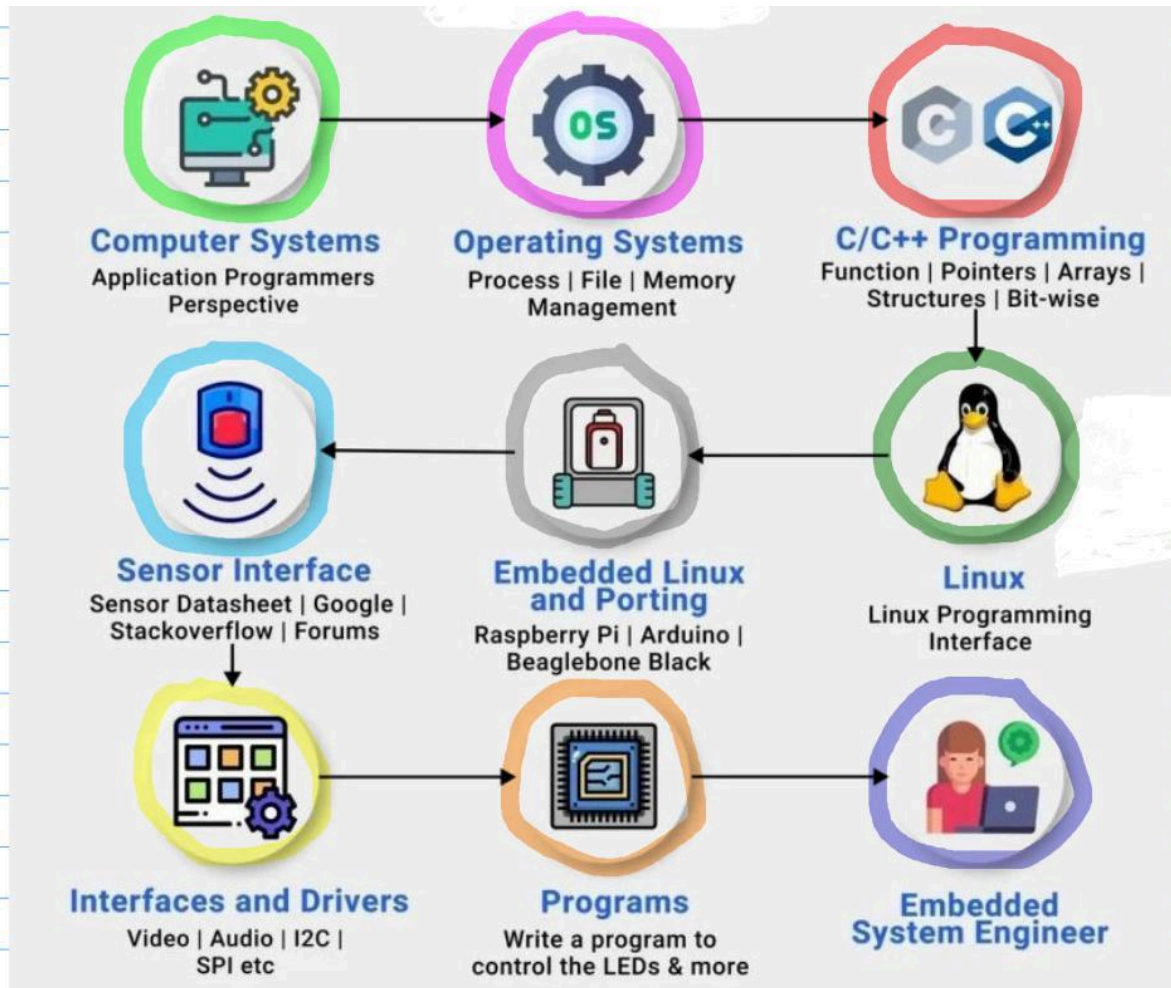
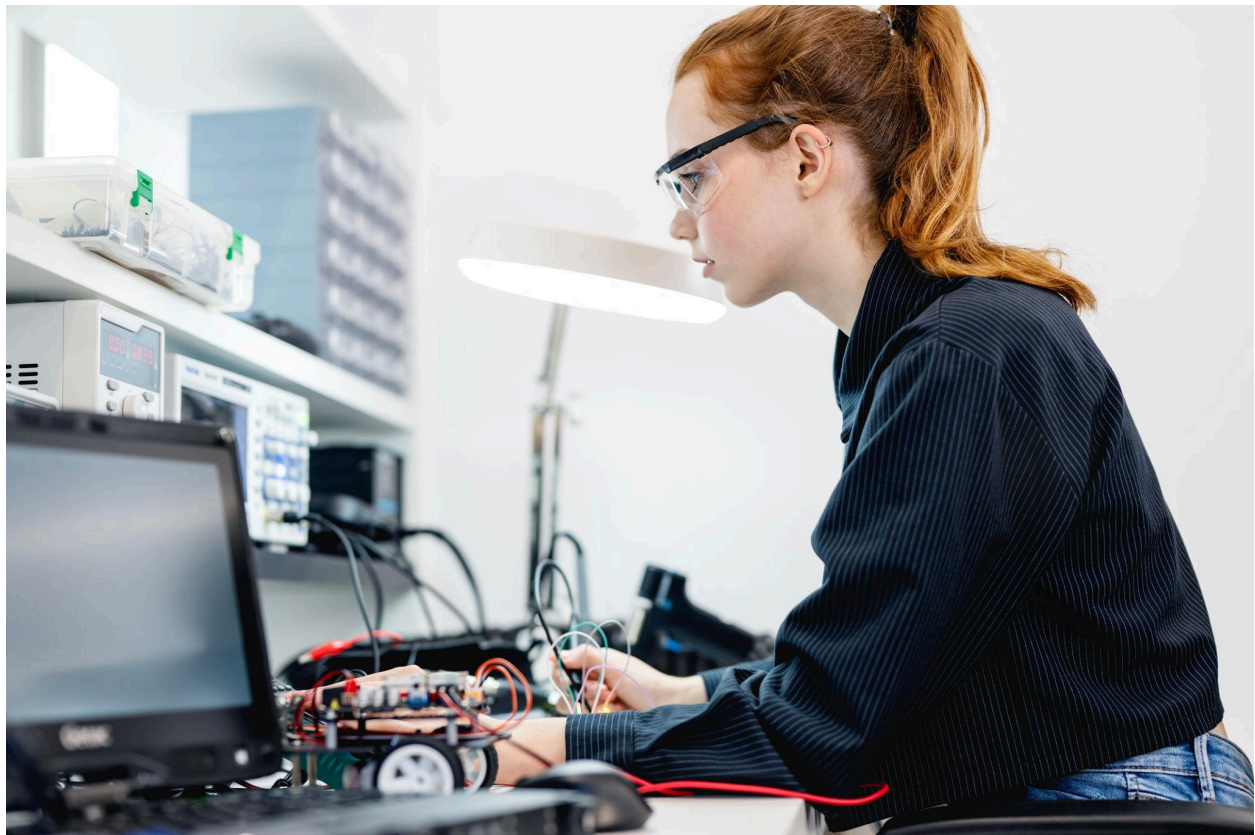
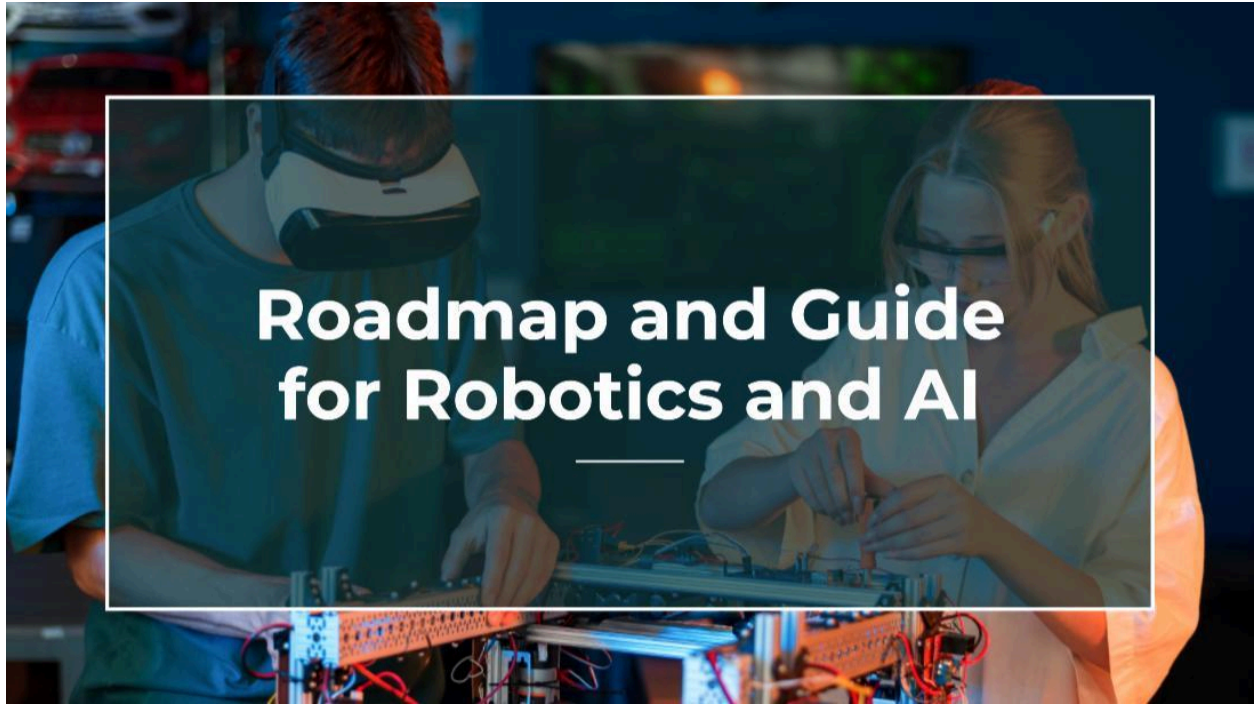


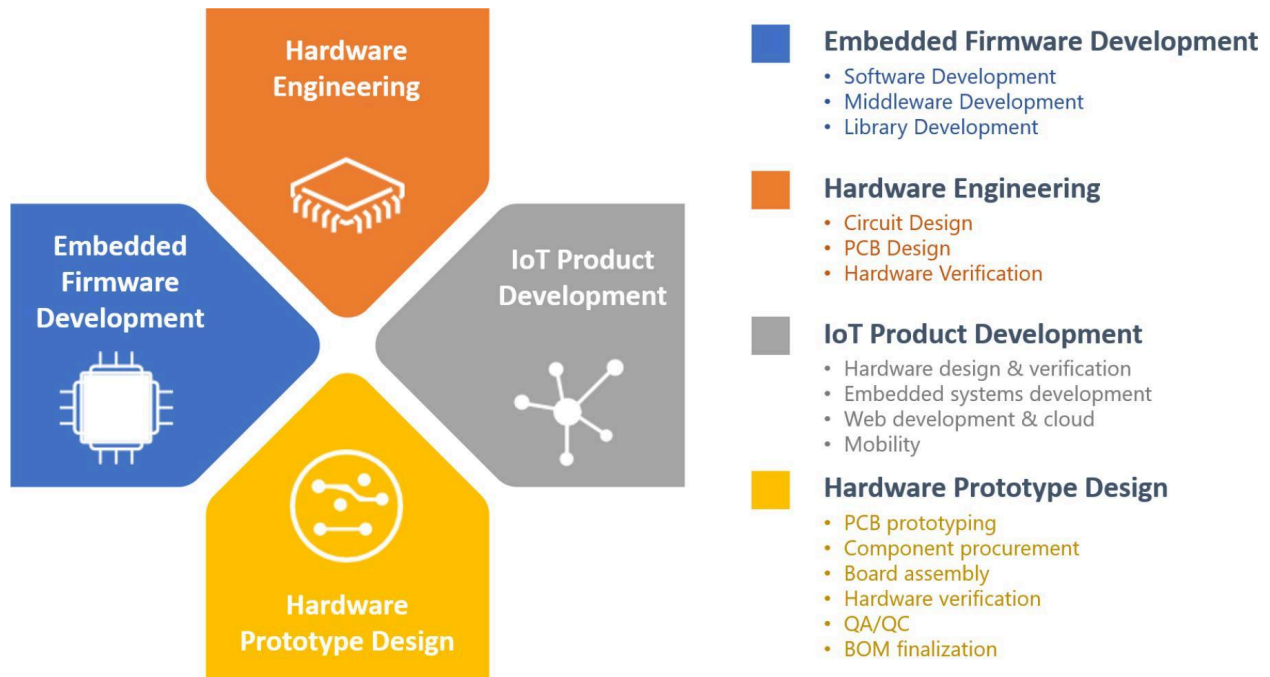
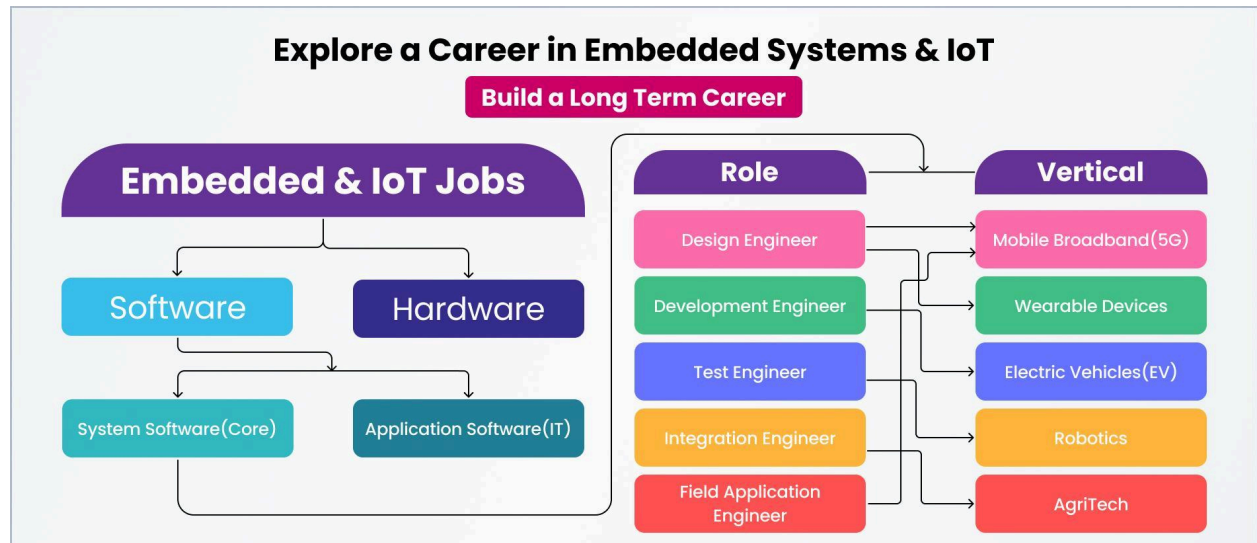
# Embedded Project Difficulty Roadmap











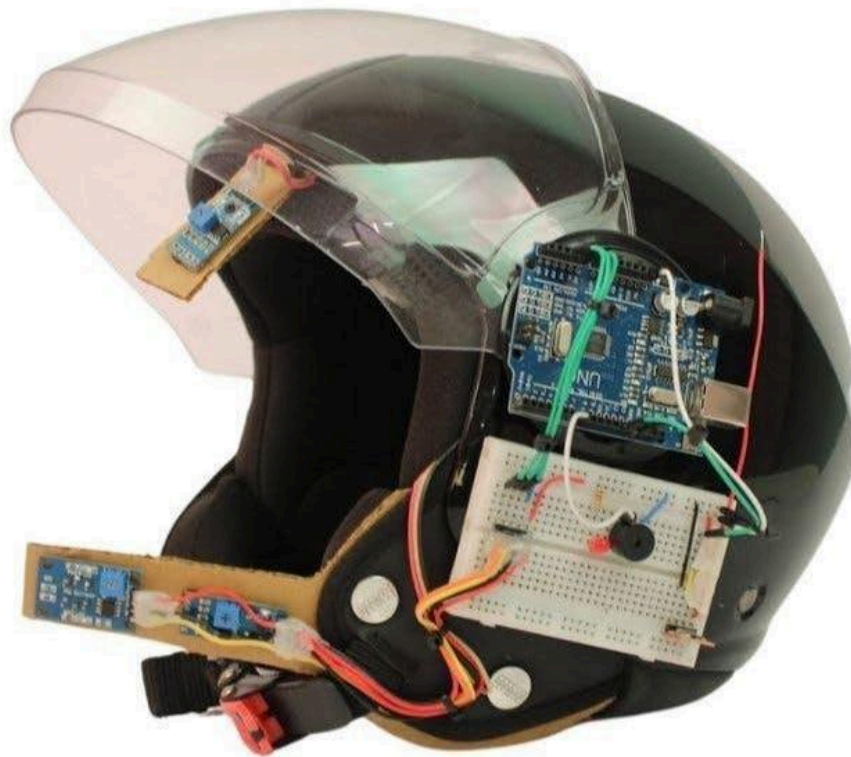
| Project            | Level        | Main Technologies   | Skills You Learn     |
|--------------------|--------------|---------------------|----------------------|
| Smart Helmet       | Beginner     | ESP32, Sensors, GSM | IoT, sensors, alerts |
| LoRa Communication | Beginner     | LoRa, ESP32         | Wireless networking  |
| Autonomous Rover   | Intermediate | Raspberry Pi, STM32 | Robotics, navigation |
| AI CCTV            | Intermediate | OpenCV, Camera, AI  | Computer vision      |

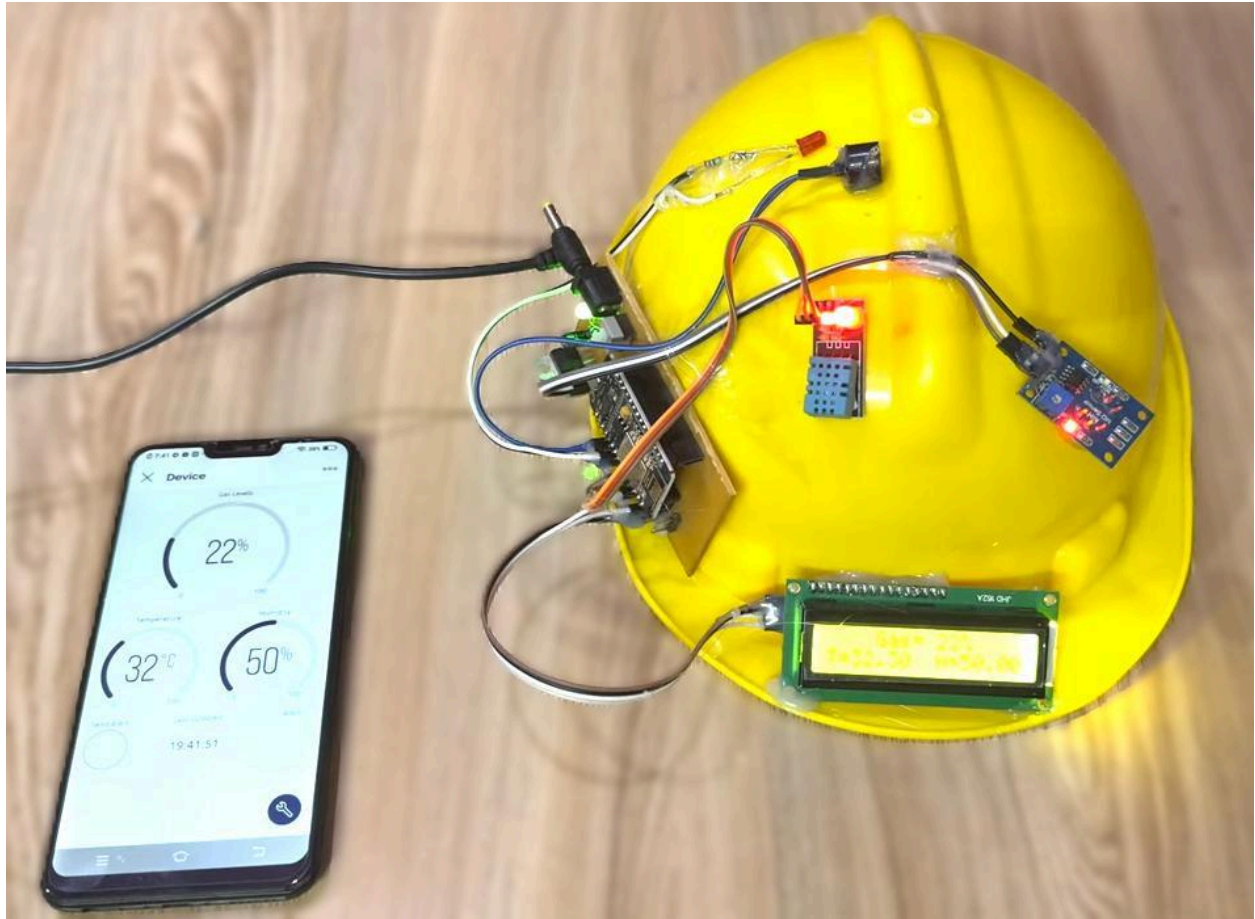


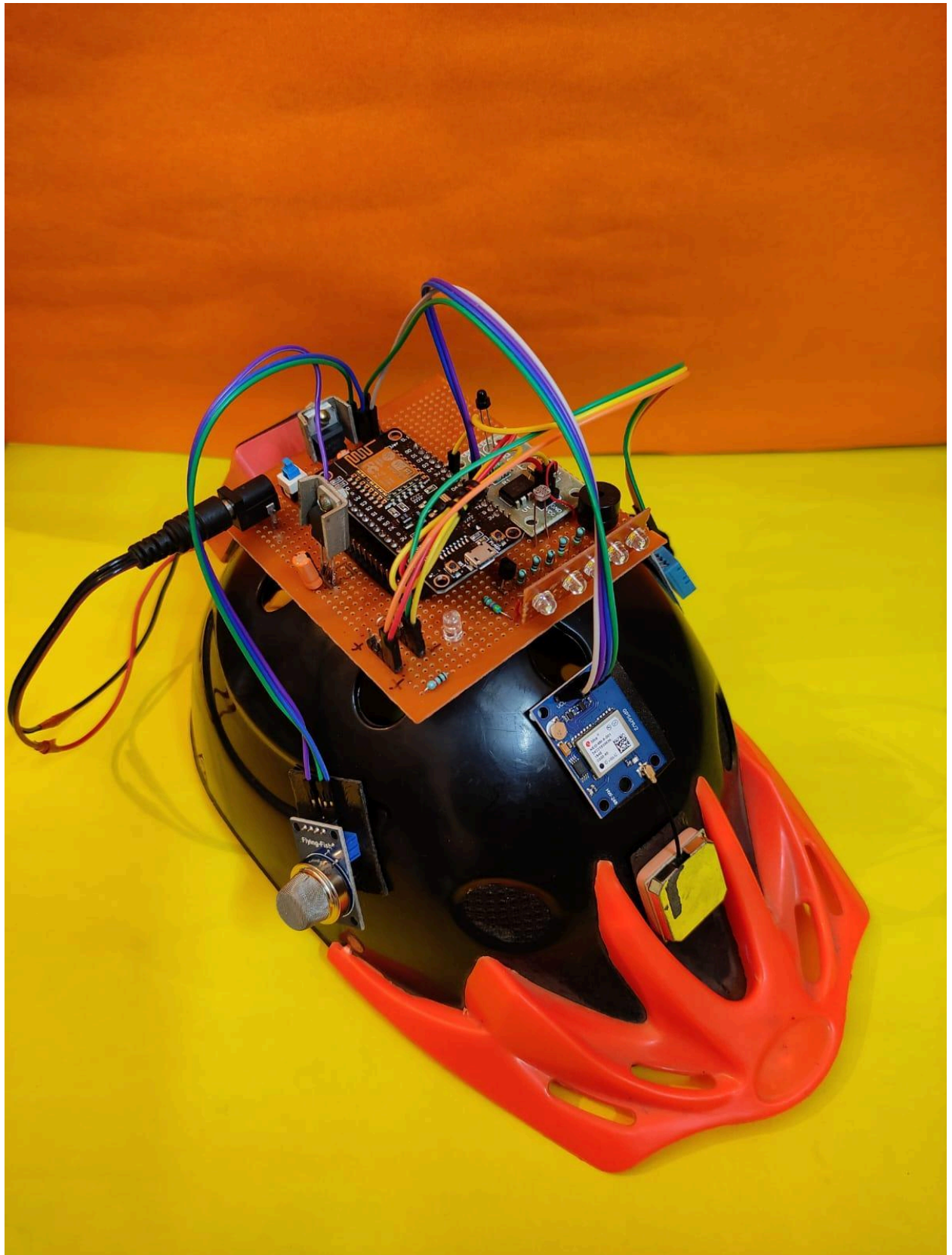
|                          |          |                      |                        |
|--------------------------|----------|----------------------|------------------------|
| Industrial AI Monitoring | Advanced | STM32, Cloud, AI     | Automation + analytics |
| Smart Defense System     | Advanced | Thermal Camera, LoRa | Surveillance systems   |
| AI Surveillance Drone    | Expert   | Jetson Nano, AI, UAV | Aerospace + autonomy   |

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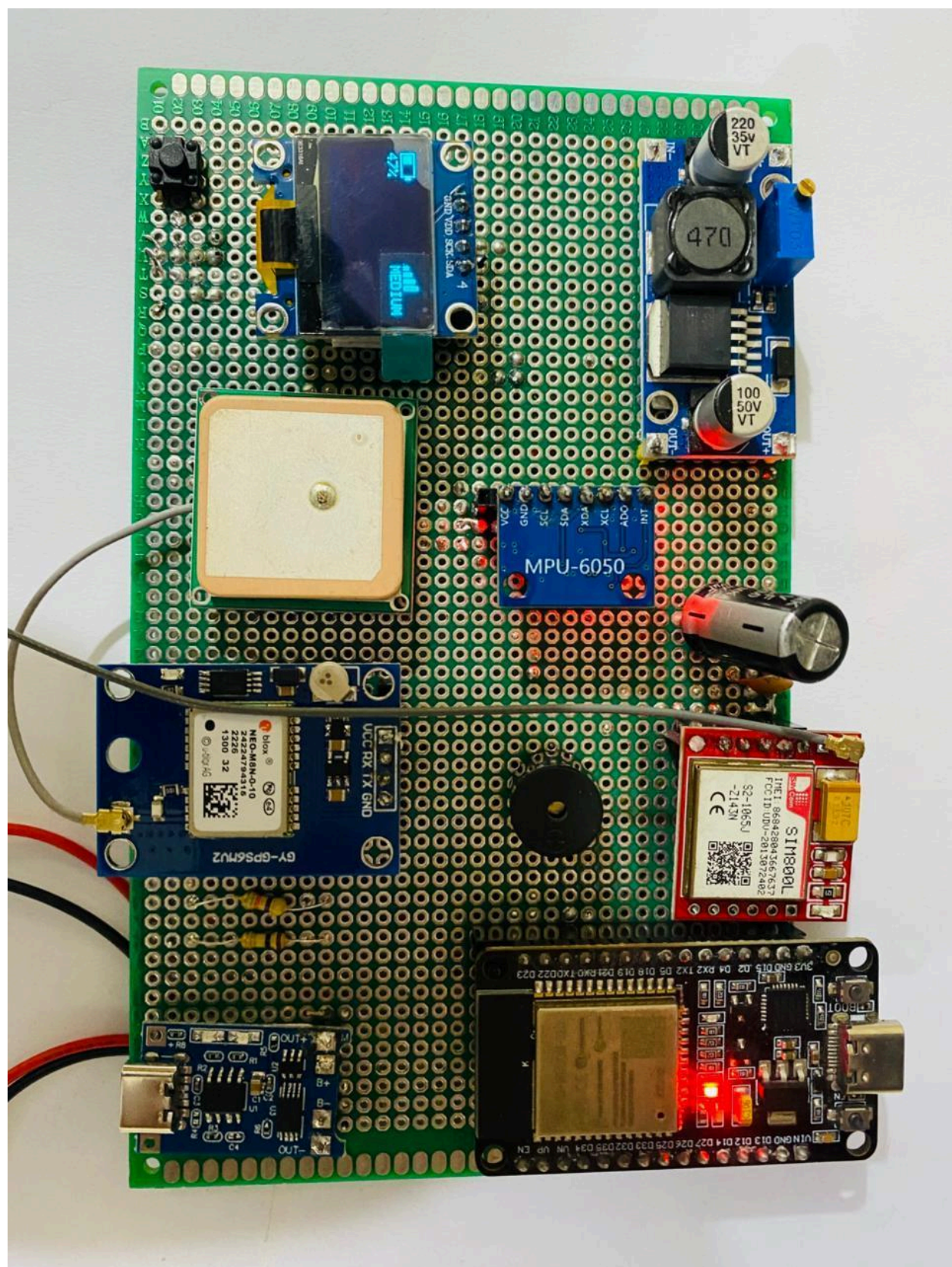
## 1. Smart Helmet (Beginner)

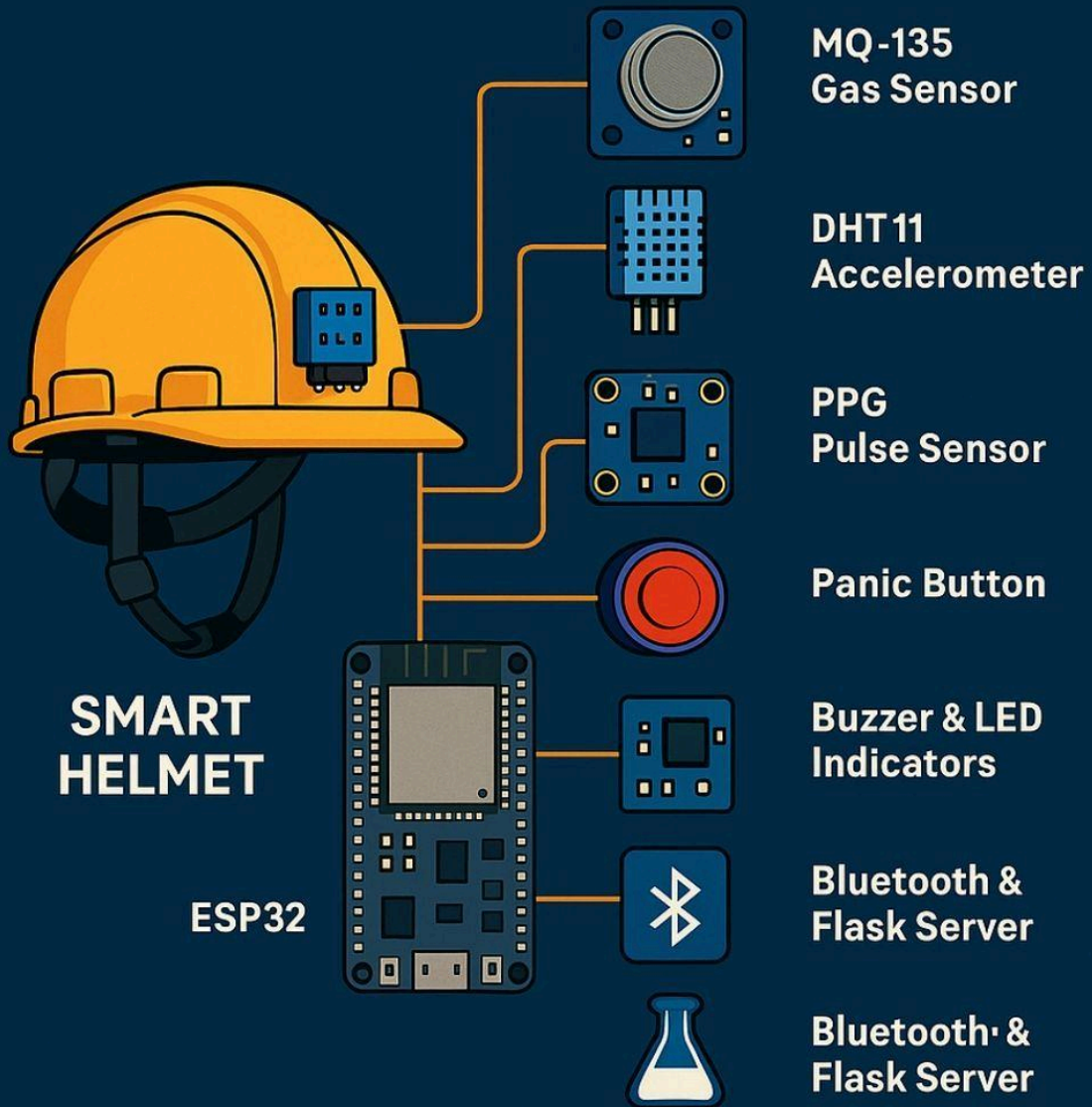


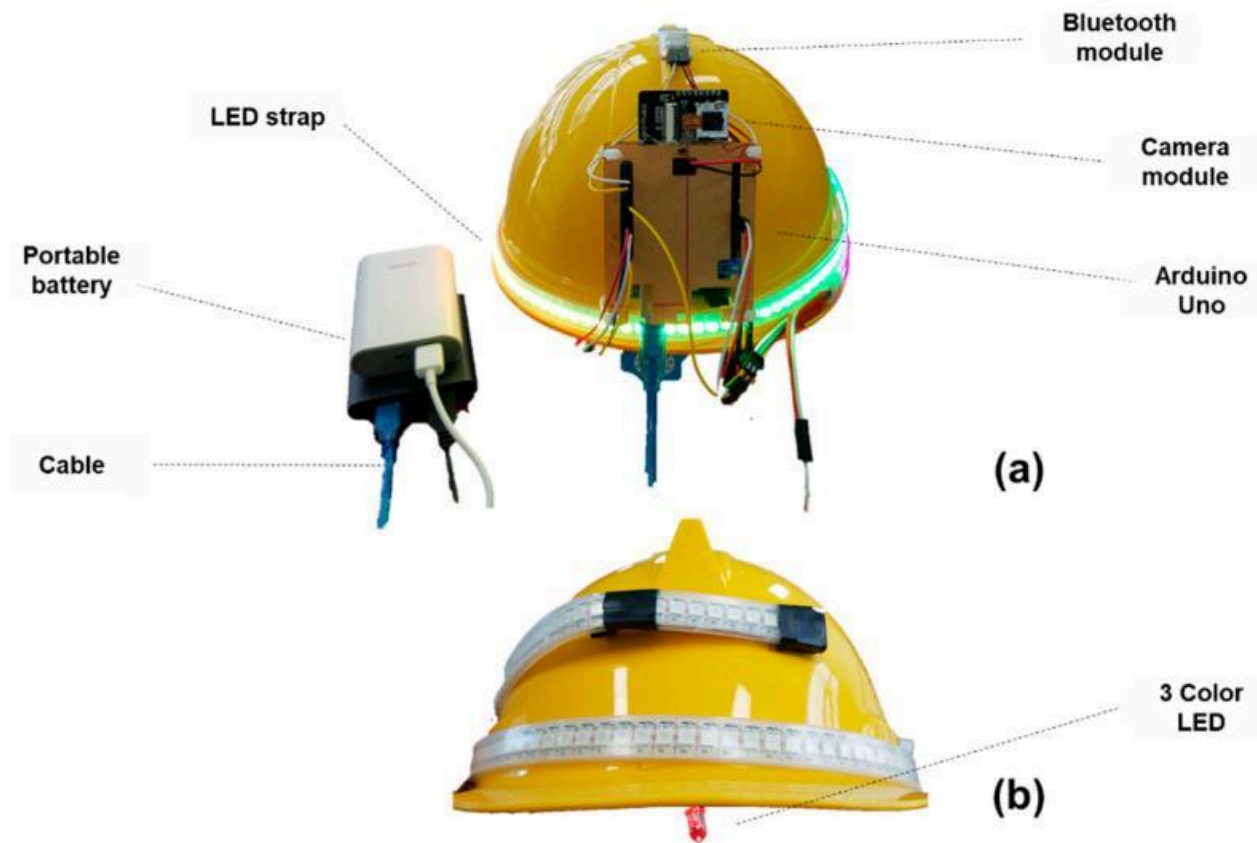












## Difficulty

★ Beginner

## Components

- ESP32
- MPU6050
- GSM Module
- GPS Module
- Alcohol Sensor

## Features

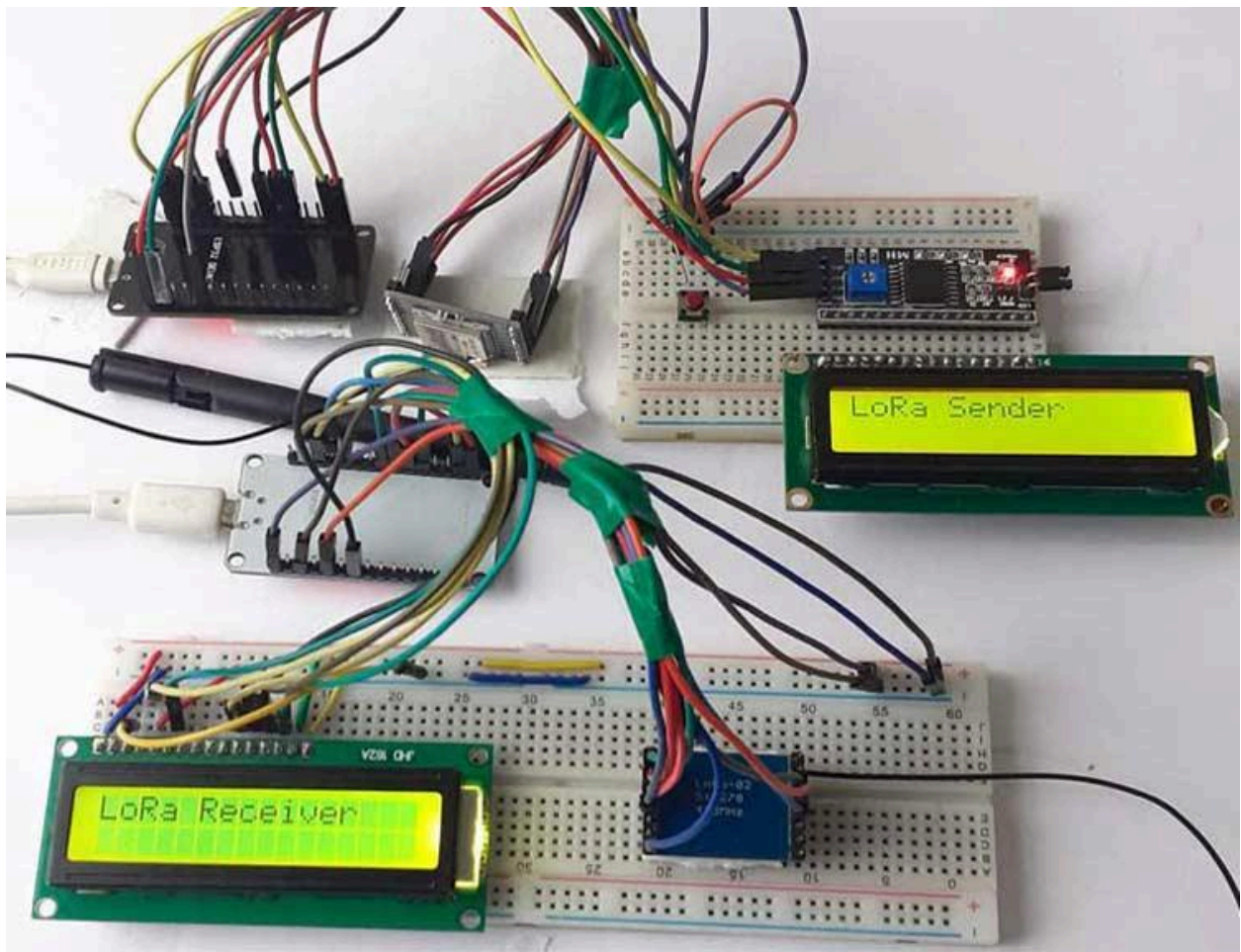
- Accident detection
- Emergency SMS
- Live location tracking
- Rider safety monitoring



## Best For Learning

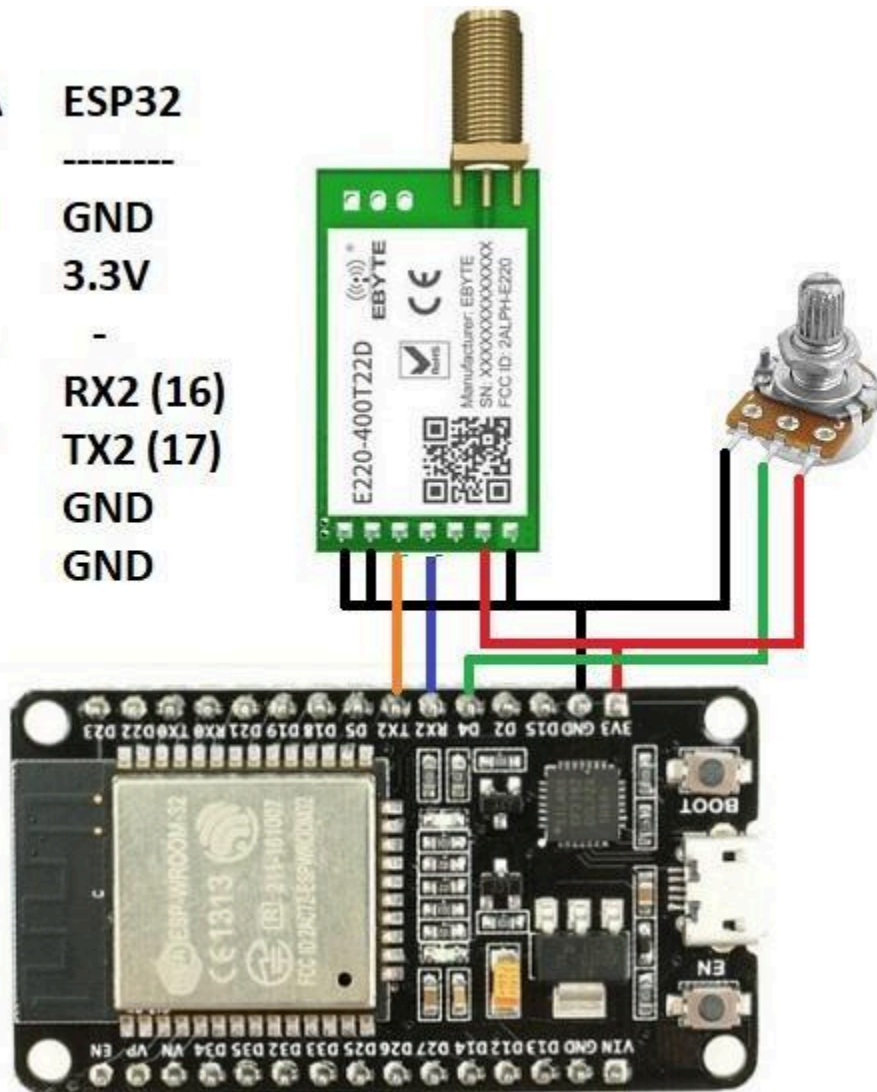
- Sensors
  - IoT
  - ESP32 programming
- 

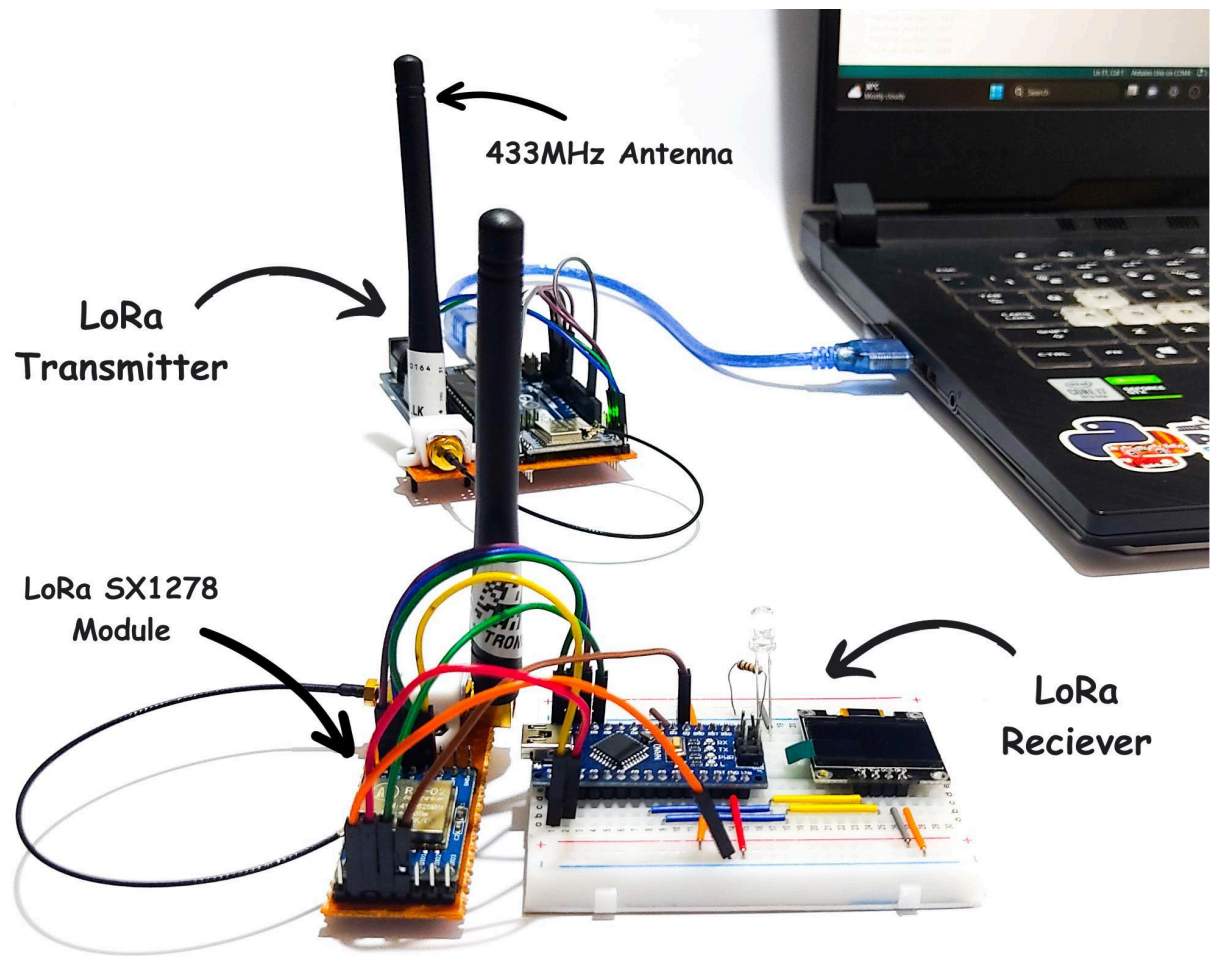
## 2. LoRa Communication (Beginner)



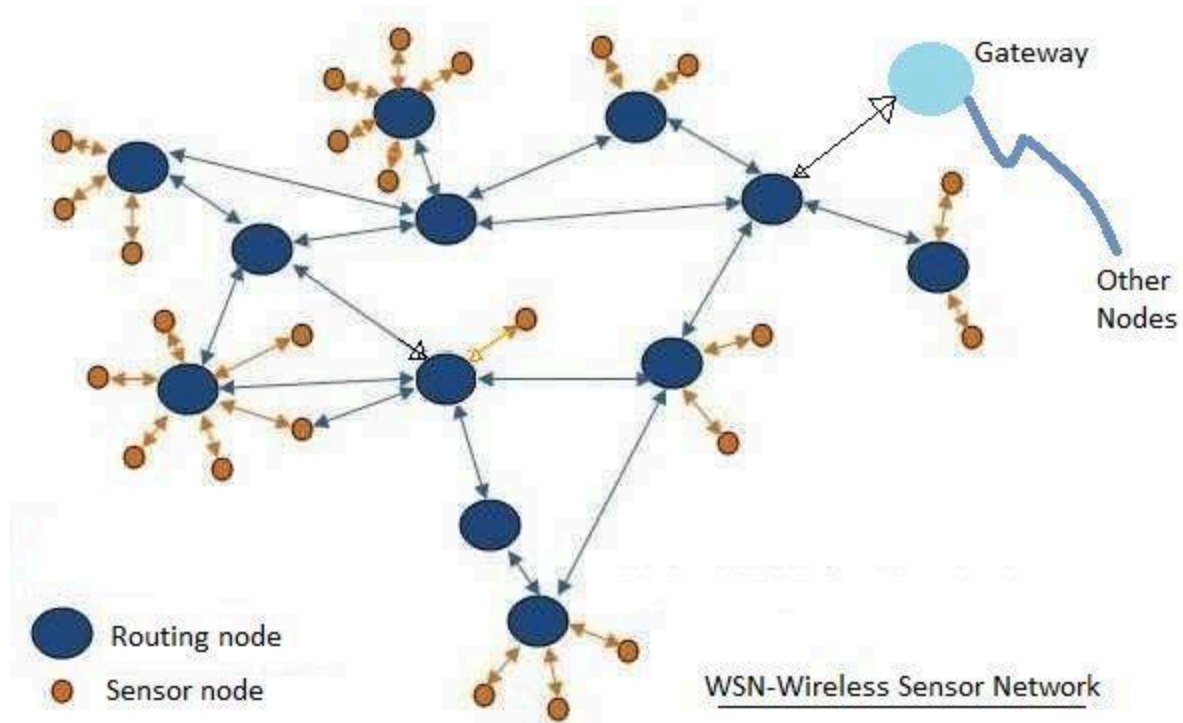


| LORA  | ESP32    |
|-------|----------|
| ----- | -----    |
| GND   | GND      |
| VCC   | 3.3V     |
| AUX   | -        |
| TXD   | RX2 (16) |
| RXD   | TX2 (17) |
| M1    | GND      |
| M0    | GND      |









## Difficulty

★ Beginner

## Components

- ESP32
- LoRa SX1278
- OLED Display

## Features

- Long-range messaging
- Sensor data transfer
- Remote communication

## Skills

- Wireless communication
- Networking basics
- IoT systems

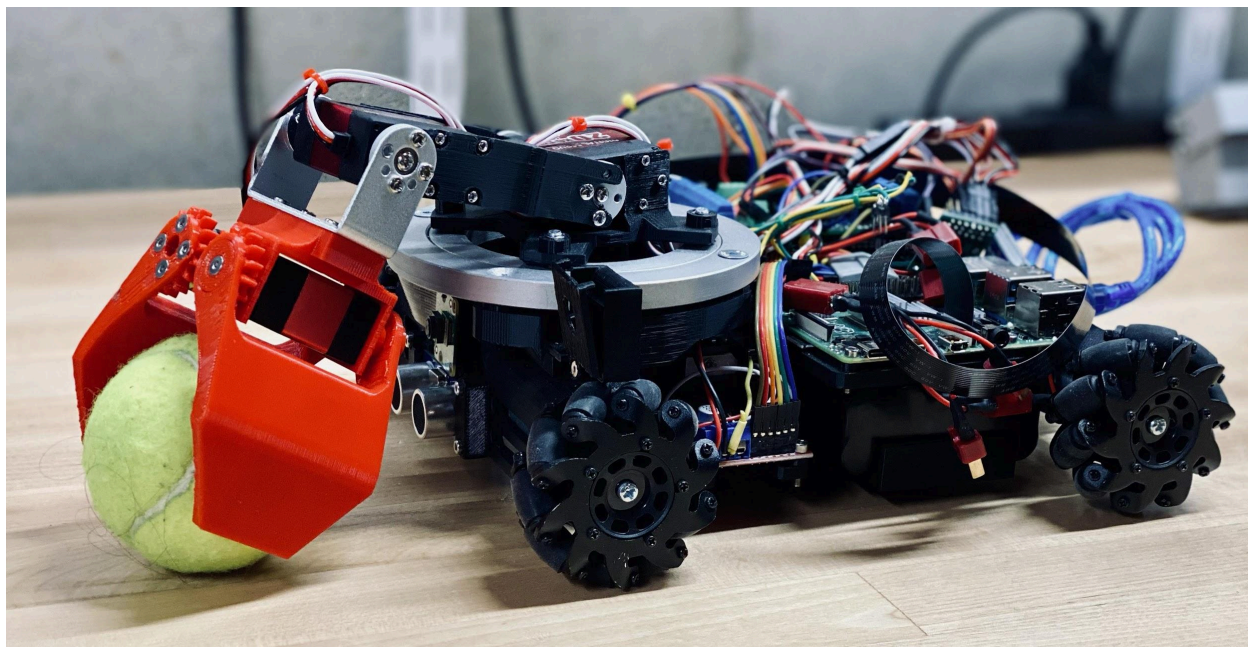
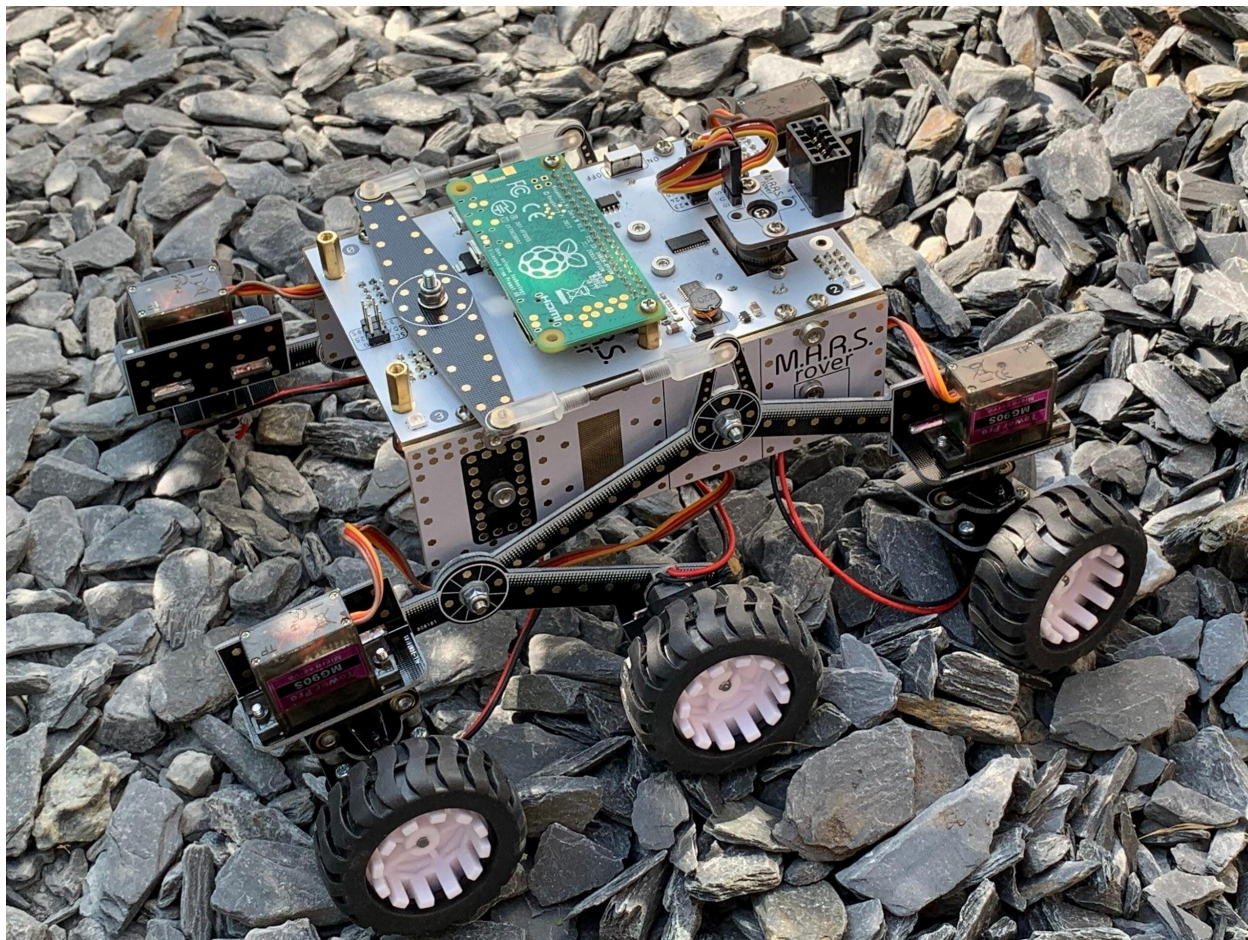
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### 3. Autonomous Rover (Intermediate)







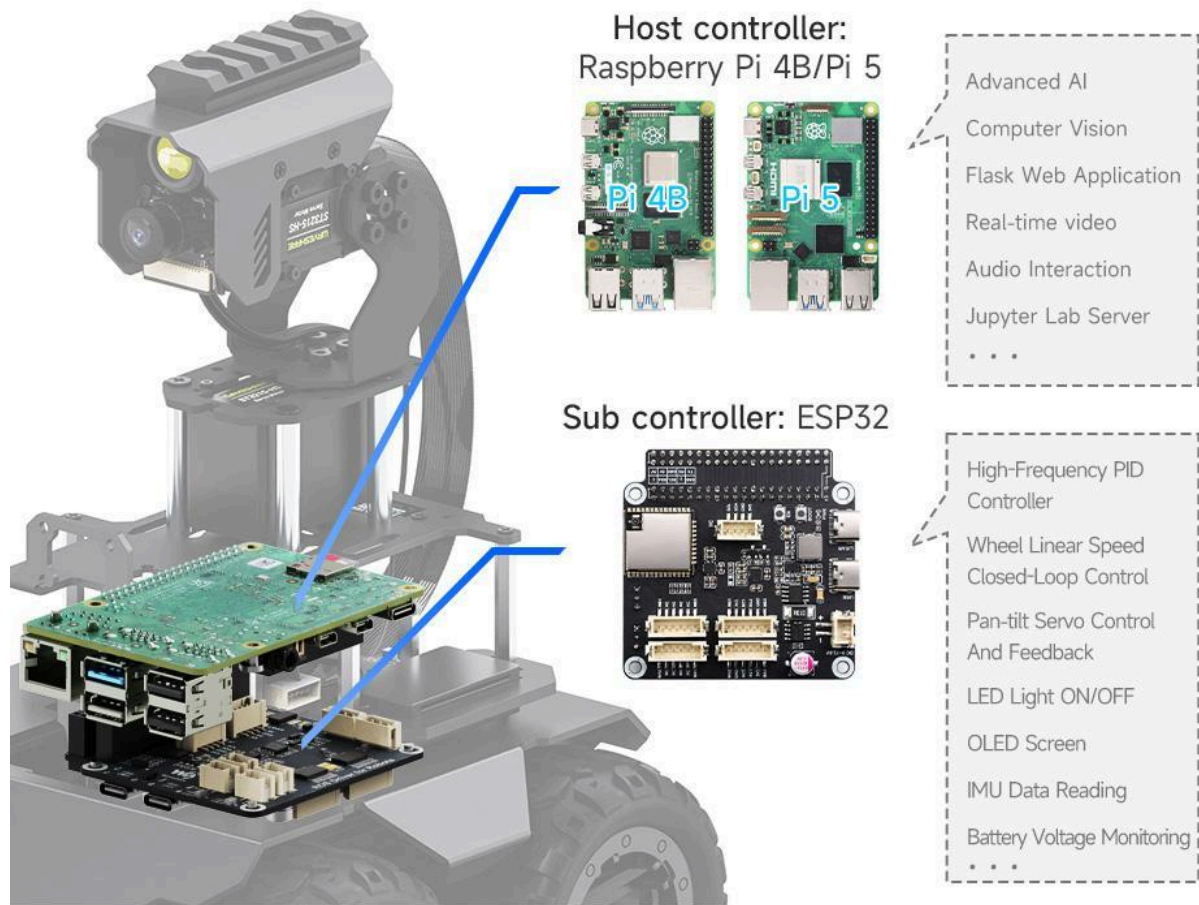








ROS



## Difficulty

★★ Intermediate

## Components

- Raspberry Pi
- STM32
- Ultrasonic Sensors
- GPS
- IMU

## Features

- Autonomous movement



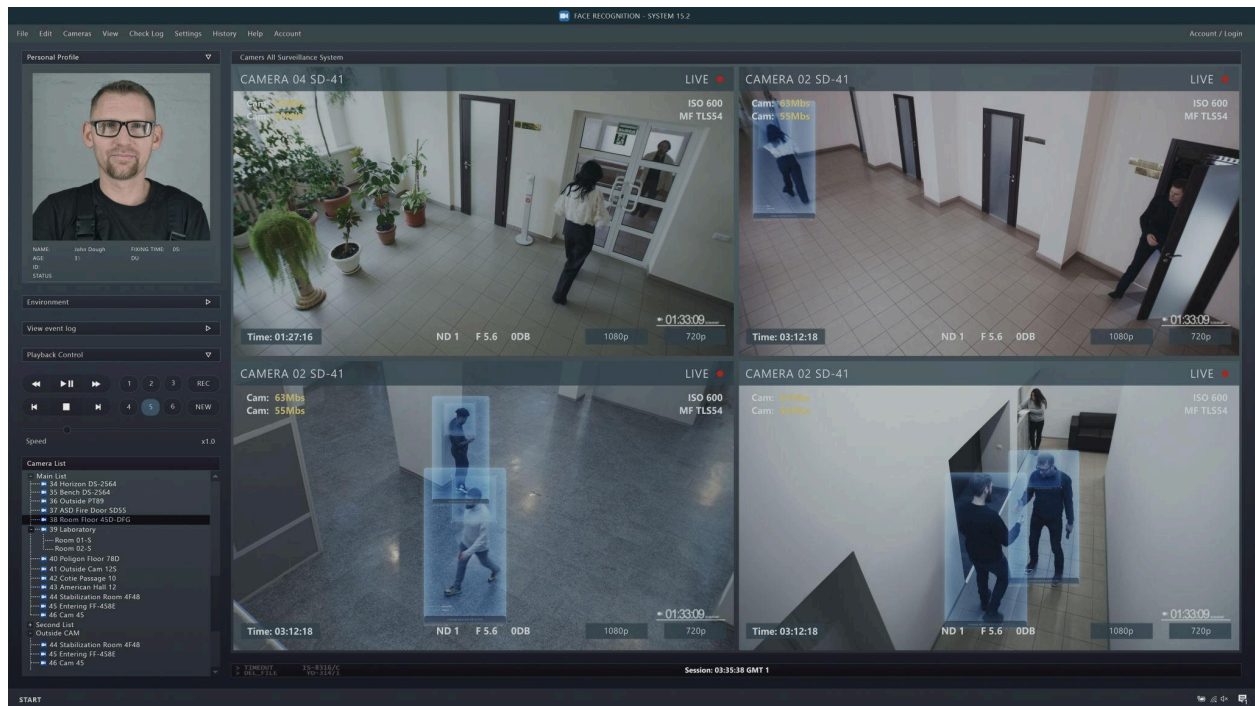
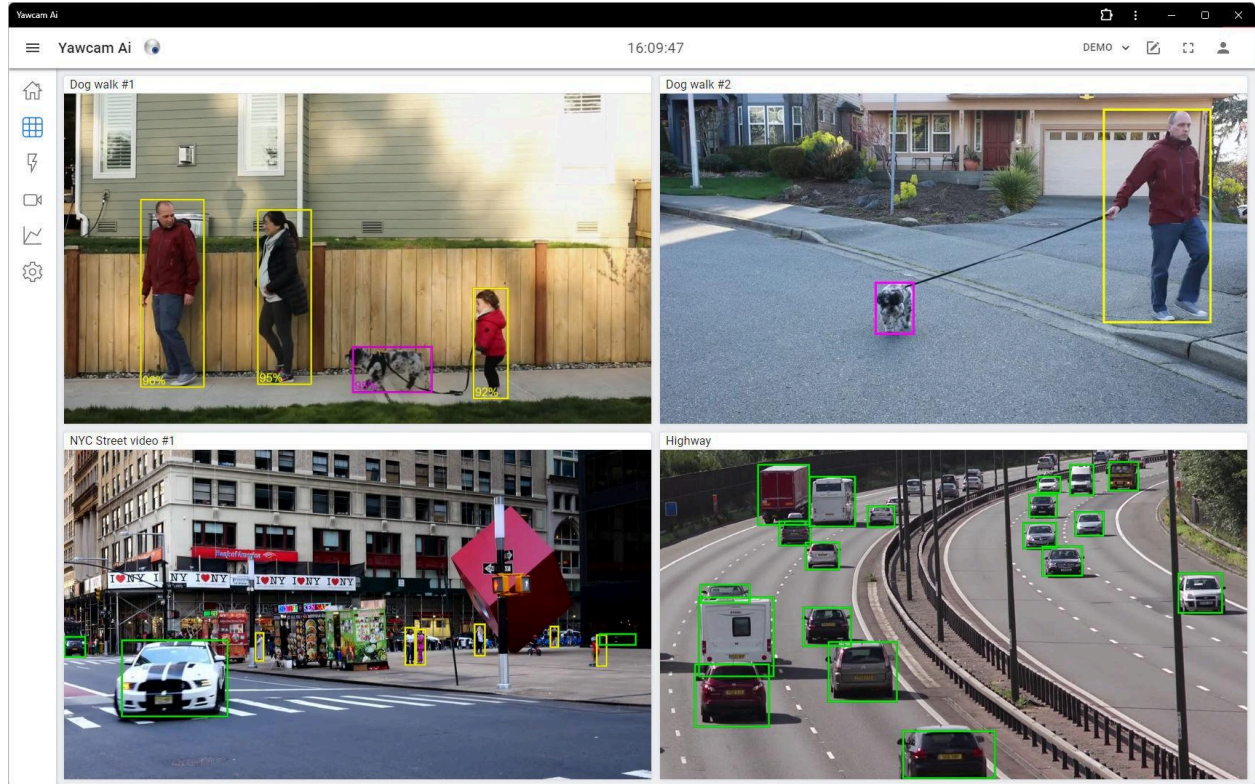
- Obstacle avoidance
- GPS navigation

## **Skills**

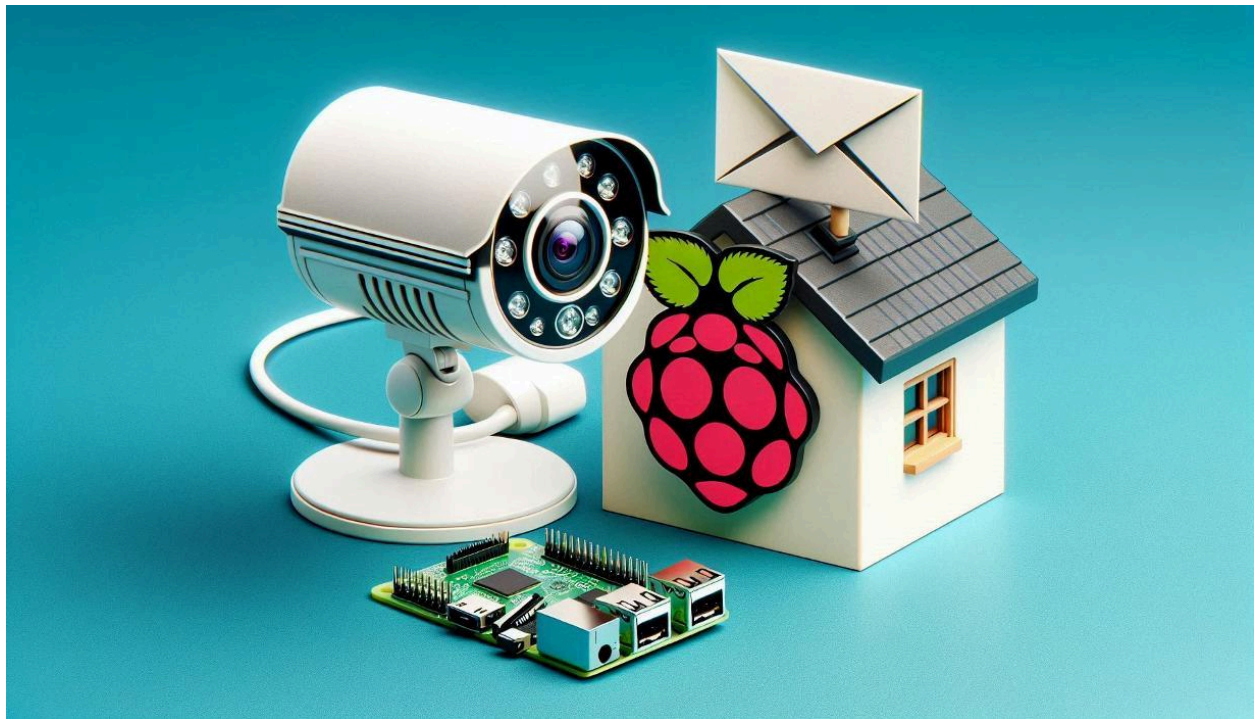
- Robotics
  - Navigation
  - Embedded Linux
- 

## **4. AI CCTV (Intermediate)**













## Difficulty

★★ Intermediate

## Components

- Raspberry Pi
- Camera Module
- OpenCV
- PIR Sensor

## Features

- Human detection
- Motion tracking
- Face recognition

## Skills

- Computer vision
  - AI basics
  - Python
- 

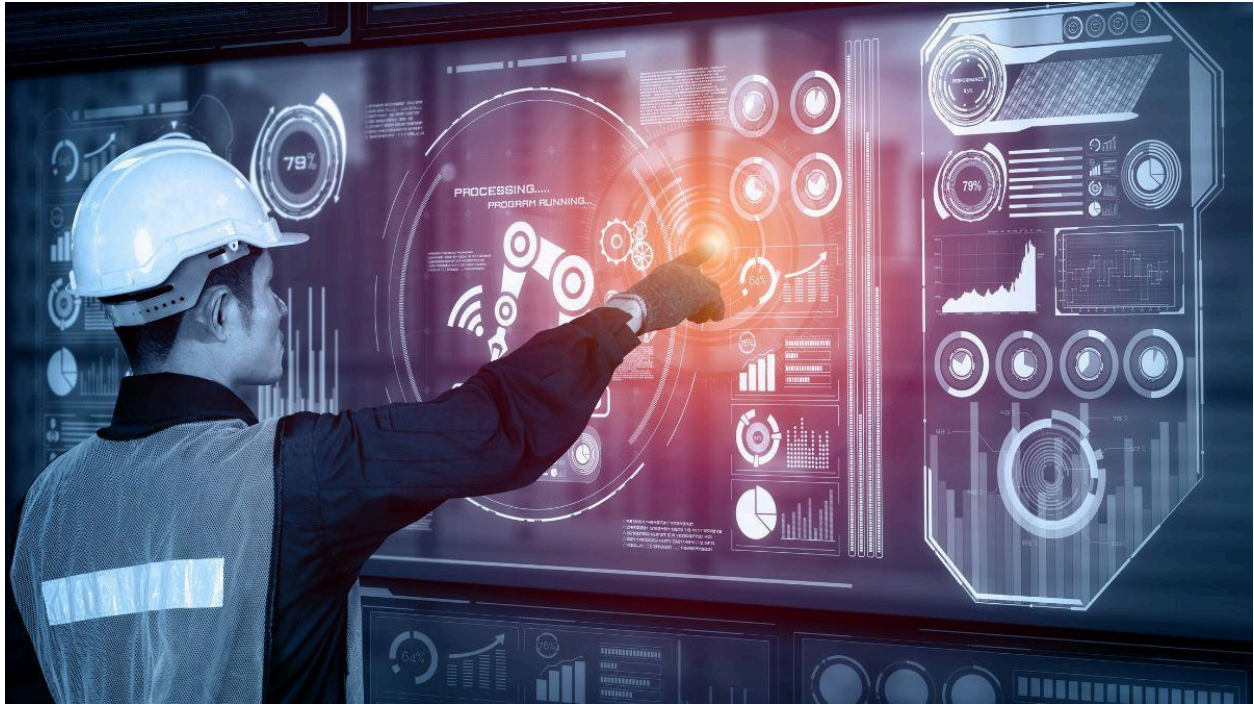
## 5. Industrial AI Monitoring (Advanced)



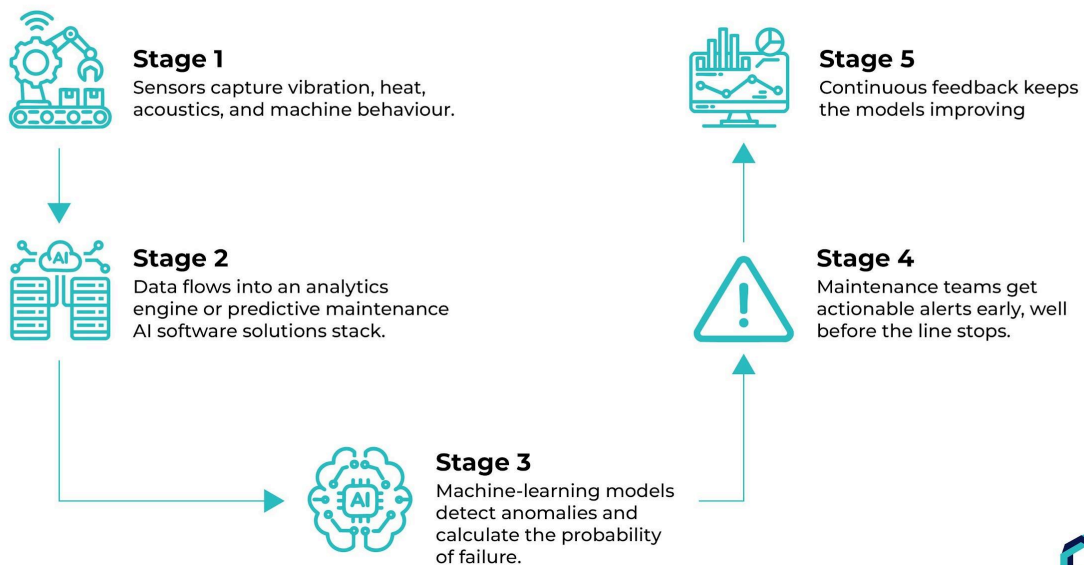








### How Do AI-Predictive Maintenance Services Work?





## Difficulty

★★★★ Advanced

## Components

- STM32
- Raspberry Pi
- Sensors
- Cloud Dashboard

## Features

- Predictive maintenance
- Real-time analytics
- Smart alerts

## Skills

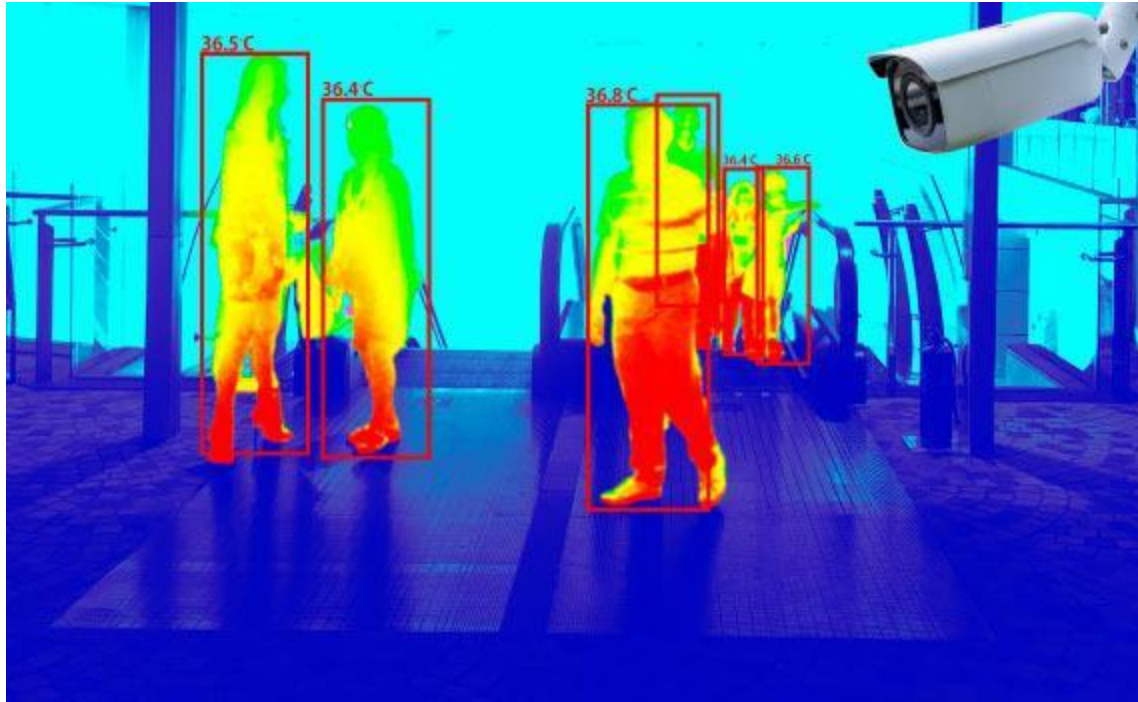
- Cloud integration
- AI analytics
- Industrial automation

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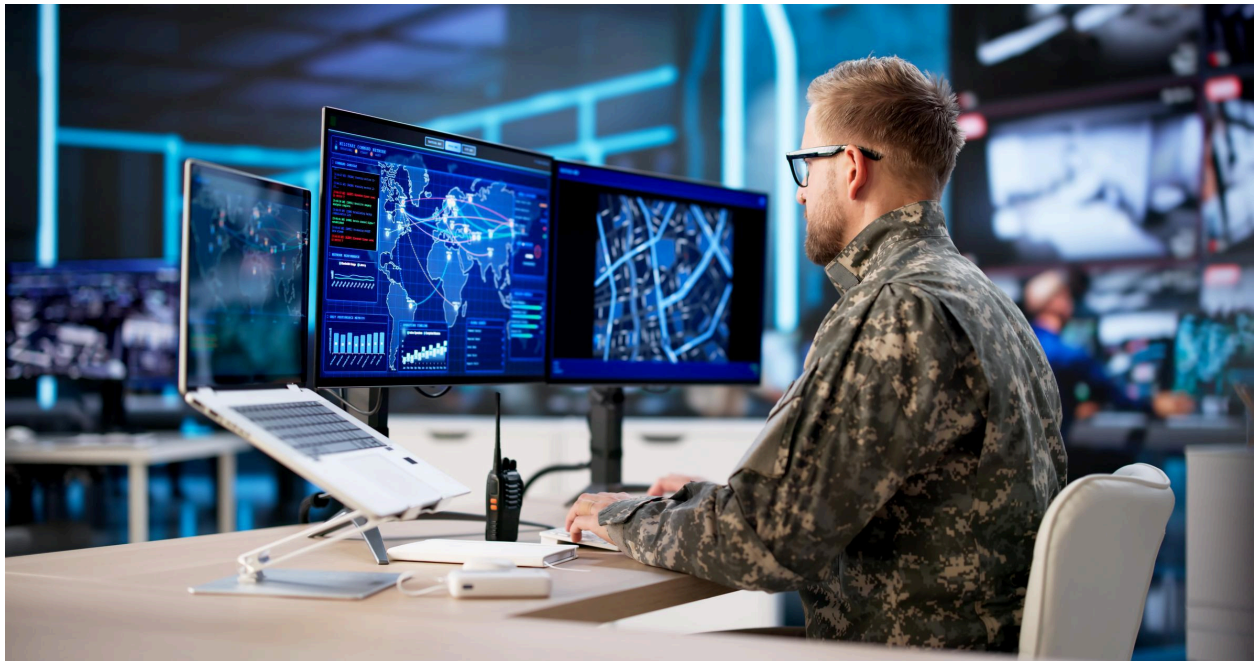
## 6. Smart Defense System (Advanced)

















## Difficulty

★★★★ Advanced

## Components

- ESP32
- Thermal Camera
- LoRa

- AI Detection

## Features

- Intruder detection
- Thermal monitoring
- Long-range wireless alerts

## Skills

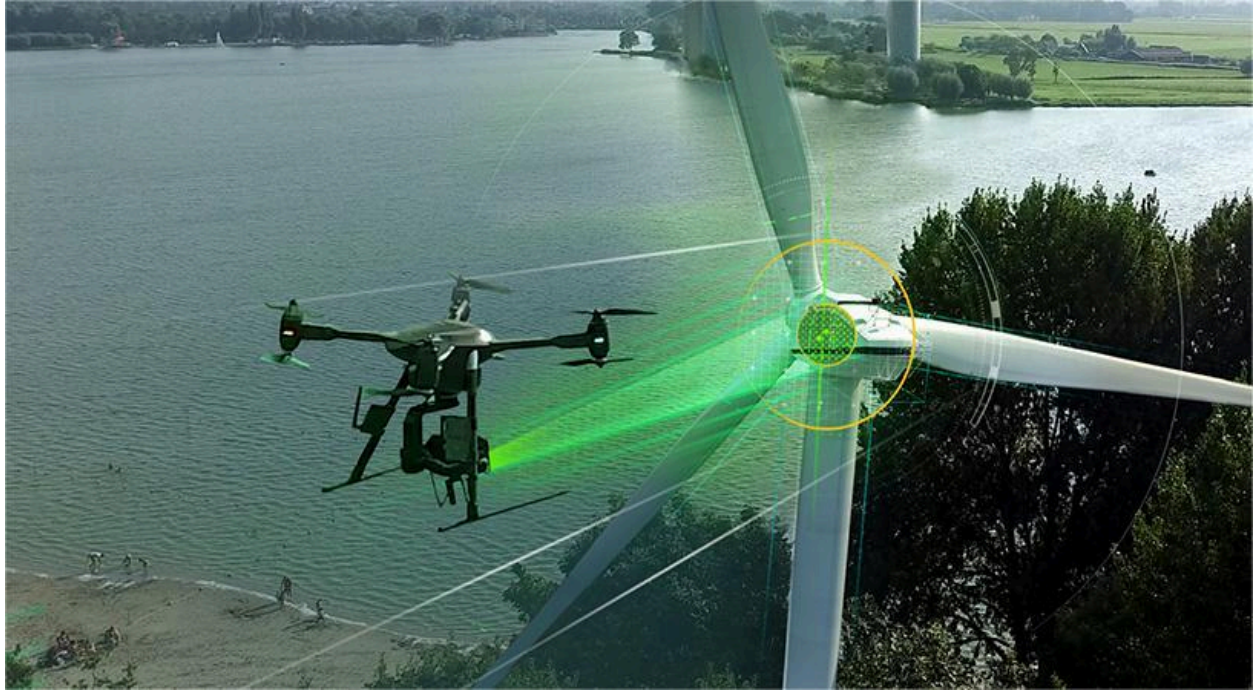
- Thermal imaging
  - AI detection
  - Defense systems
- 

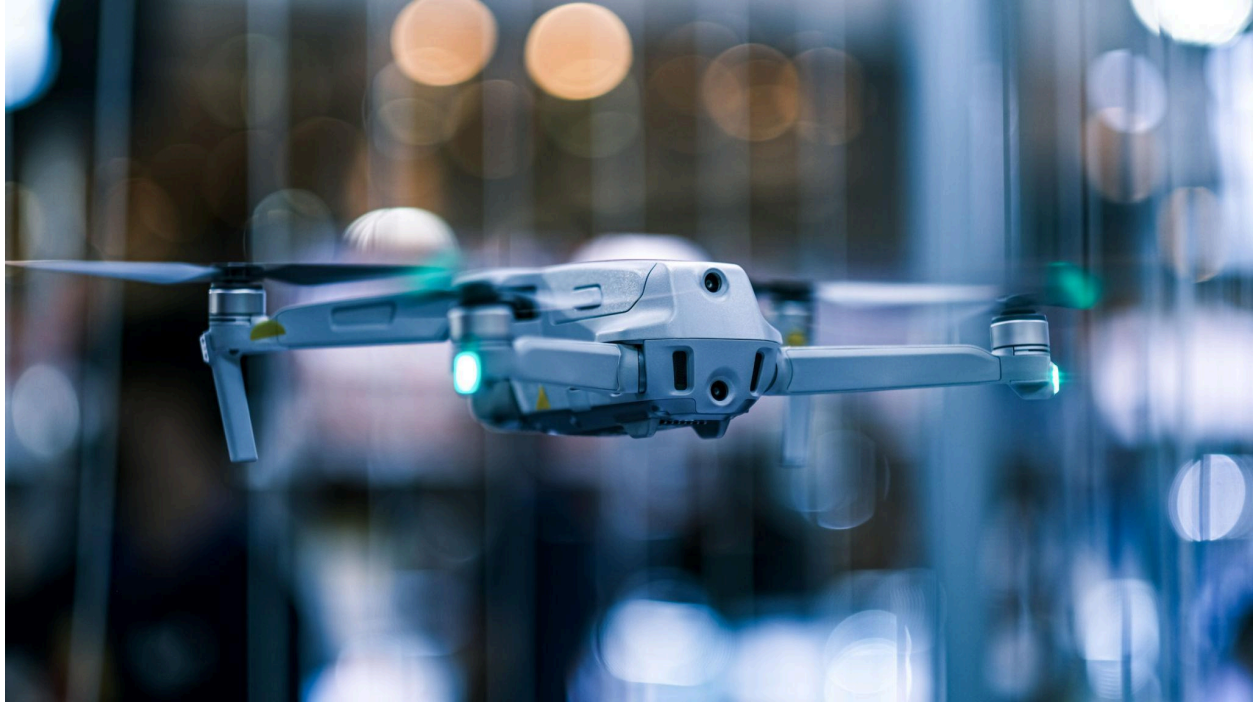
# 7. AI Surveillance Drone (Expert)













## Difficulty

★★★★★ Expert

## Components

- Jetson Nano
- Thermal Camera
- GPS
- IMU
- LoRa



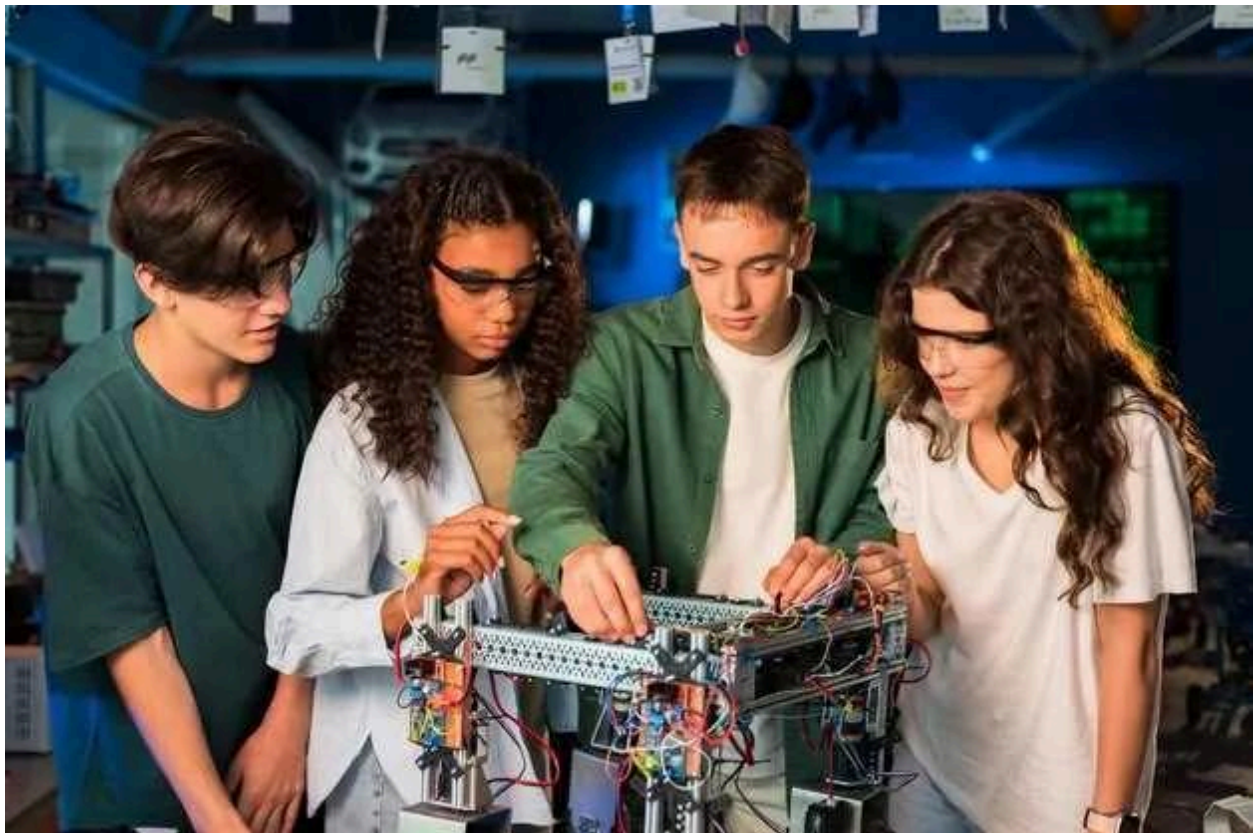
## Features

- Autonomous flight
- AI tracking
- Thermal human detection
- Live surveillance

## Skills

- Aerospace systems
  - Drone AI
  - Autonomous robotics
  - Advanced networking
- 

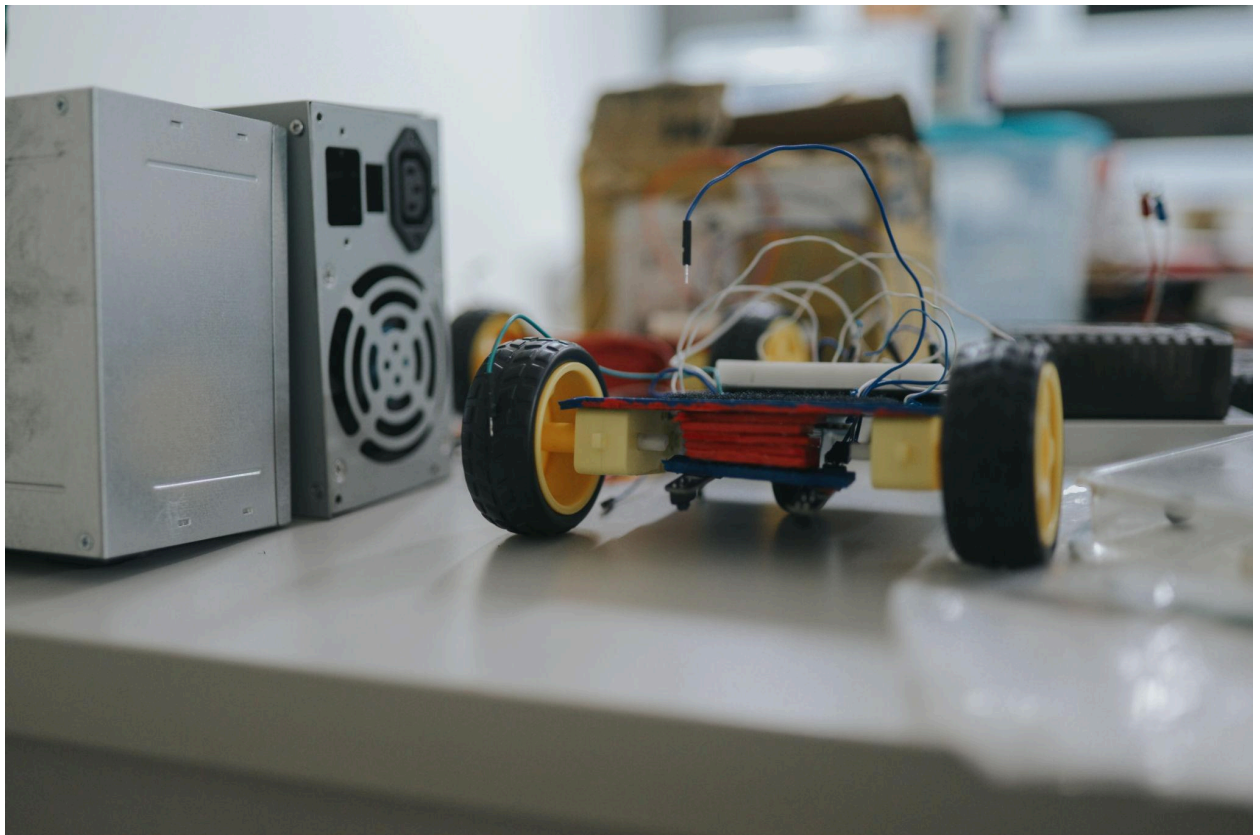
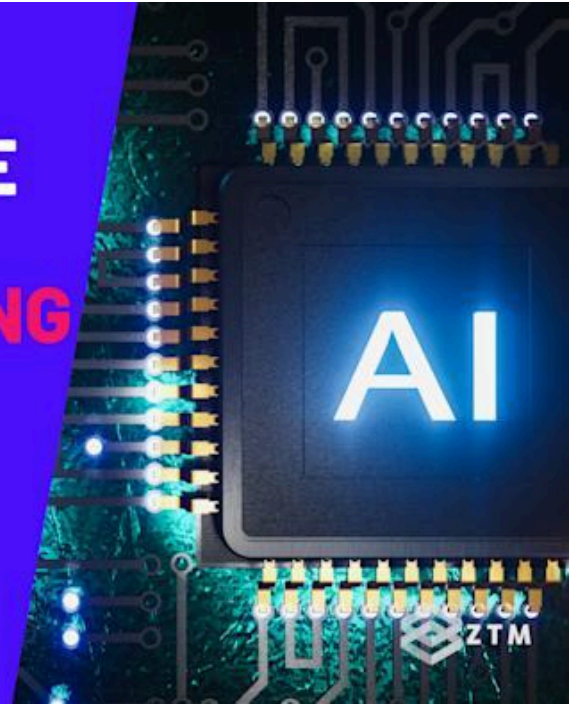
## Suggested Learning Timeline



# BEGINNER'S GUIDE TO **EMBEDDED** **MACHINE LEARNING**



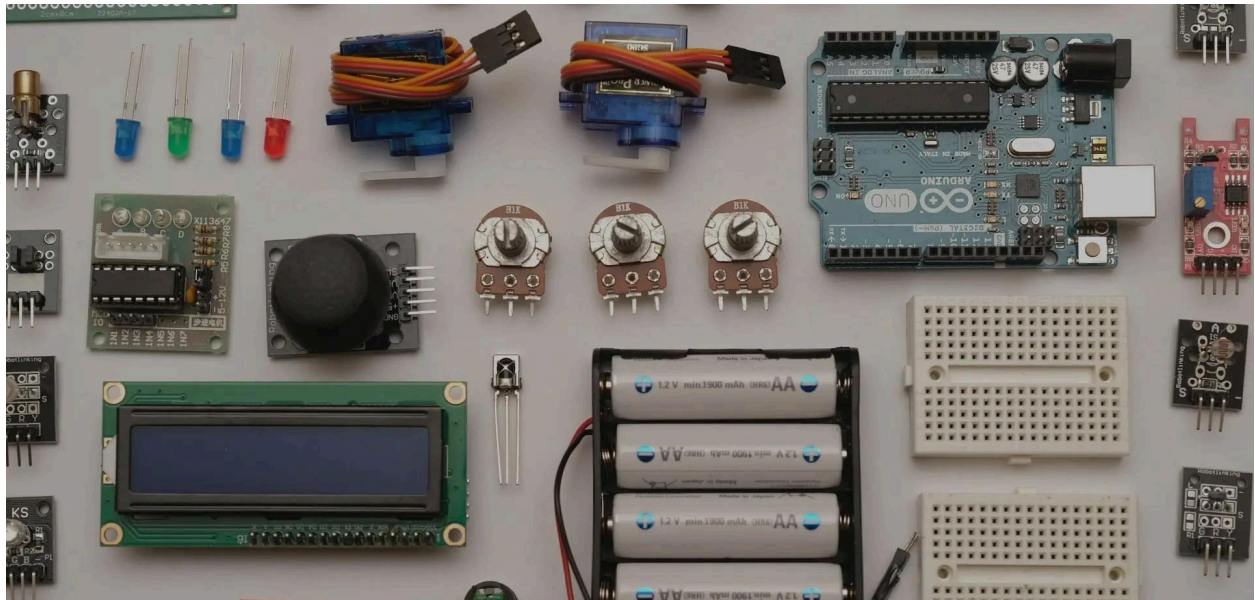
Written by  
Daniel Bourke











| Phase   | Duration   | Focus                |
|---------|------------|----------------------|
| Phase 1 | 1–2 Months | Electronics + ESP32  |
| Phase 2 | 2–3 Months | Raspberry Pi + Linux |
| Phase 3 | 2 Months   | LoRa + Networking    |
| Phase 4 | 3 Months   | OpenCV + AI          |
| Phase 5 | 3–6 Months | Robotics + Drones    |

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